Bunyan Yorganci

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EDUCATION

Carleton University — B.Eng., Computer Systems Engineering **Sep 2022 – Apr 2028**

Relevant coursework: Operating Systems, Real-Time Concurrent Systems, Real-Time Systems, Computer Systems Design, Computer Organization and Architecture, Algorithms and Data Structures, Software Validation, Computer Communications

TECHNICAL SKILLS

- **Languages:** C, C++, Python, Java, Verilog, ARM assembly, MATLAB, Bash
- **Embedded:** Arduino Due, Raspberry Pi, TI MSP432, Xilinx Zynq; GPIO, PWM, interrupts, timers, ADC, UART, I2C, SPI, DMA, AXI; motor drivers, soldering
- **Systems:** Linux, POSIX IPC (fork, shared memory, semaphores), C++ threads and synchronization, UDP sockets, state machines, real-time scheduling
- **Testing and tools:** JUnit 5, Gradle, JaCoCo, mutation and model-based testing, Git, Vivado, Wireshark, MATLAB/Simulink, Flask, OpenCV, SQLite, Docker
- **Certification:** AWS Certified Cloud Practitioner

PROJECTS

Autonomous Snowplow Robot — SYSC 4805 **Fall 2026 (in progress)**

- Building, in a team of four, an Arduino Due robot that must clear simulated snow from a taped arena in 5 minutes while avoiding fixed and moving obstacles
- Developing the motor-driver direction and stop routines for the four-motor platform, exposed as forward, reverse, turn and stop commands for team testing
- Integrating rear-wheel encoder feedback into the team's state-machine controller, which combines IMU, line-following and distance sensors

Multi-threaded Elevator Control System — SYSC 3303 **Winter 2025**

- Built in C++ a scheduler, elevator simulator and floor simulator as three processes communicating over UDP, able to run on separate machines
- Implemented a multi-threaded scheduler handling several cars at once, with configurable floors, elevators, door timing and travel time

Operating-System and Interrupt Simulator — SYSC 4001 **2024–25**

- Implemented in C++ on Linux a trace-driven simulation of CPU events, interrupts and system calls with process-control-block state transitions
- Modelled FORK/EXEC, fixed-partition memory allocation and scheduling, and analyzed execution traces and I/O scenarios

Linux Process Synchronization — SYSC 4001 **Fall 2025**

- Wrote multiprocess C programs using fork(), shared memory and semaphores to coordinate access to shared data, and documented a livelock found in testing

Software Validation and Boundary Testing — SYSC 4101 **Fall 2026**

- Designed parameterized JUnit 5 boundary tests reaching 100% instruction and branch coverage of a board-bounds method, with JaCoCo reports generated through Gradle

Raspberry Pi Hardware Control — SYSC 3010 **Winter 2025**

- Programmed a traffic-light and pedestrian-button controller in Python on GPIO, using assertions to verify hardware states
- Built browser-to-Sense-HAT LED controls with Flask-SocketIO

LAB EXPERIENCE

- **Xilinx Zynq SoC (SYSC 3320):** built circuits in Vivado, then wrote bare-metal C and assembly on the ARM core for interrupt-driven GPIO, AXI timers and UART
- **TI MSP432 (SYSC 3310):** programmed digital and analog I/O, interrupts and timers in C; studied race conditions and mutex synchronization
- **Signals and networks (SYSC 3501, 4602):** simulated BPSK/QPSK modulation and SNR effects; analyzed Ethernet, 802.11 and IPv4 traffic in Wireshark

EXPERIENCE

Computer Refurbisher — Self-employed, Milton, ON **2017 – Present**

- Diagnose and repair used laptops: replace screens, batteries, keyboards and I/O controllers, upgrade RAM and storage, and repair solder joints and fuses
- Clone and recover OS installations without data loss; install Windows, Linux and dual-boot systems for personal use and resale

Home Lab — Personal project **Ongoing**

- Run an Ubuntu server with Docker (Jellyfin, network storage, WireGuard VPN) and Pi-hole DNS filtering on a Raspberry Pi

VOLUNTEERING

Community Volunteer — Turkish Community, Milton and Edmonton **2017 – Present**

- Plan and run community events and cultural festivals

Cadet — Canadian Armed Forces Cadet Program, Edmonton **2019 – 2020**

- Completed the public speaking program and helped run a recycling fundraiser